

Hafeez Khan

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EDUCATION

- Florida Institute of Technology** Florida, USA
• *Masters & Ph.D. in Computer Science (GPA: 3.84, PhD Advisor: Prof. Siddhartha Bhattacharyya)* Expected 2027
- Birla Institute of Technology and Science, Pilani** Dubai, UAE
• *Bachelors of Engineering in Computer Science (GPA: 3.70, Thesis Advisor: Prof. Raja Muthalaghu)* 2023

RESEARCH AND WORK EXPERIENCE

- Research Assistant at ASSIST Lab (Florida Tech)** Fall 2023 – Present
- NASA-Funded Research** Fall 2024 – Present
Guide: Natasha Neogi, Sarah Lehman, Siddhartha Bhattacharyya **NASA Langley Research Center, USA**
 - Topics: Few Shot Learning, Self-Supervised Learning, Object Detection, Knowledge Distillation, Generative AI
 - Developed multi-modality (camera + lidar) perception algorithms for automated labeling of over **300,000** lanes and taxiways images, improving detection accuracy by **8.04%** and reducing latency per frame by **58.38%**.
 - Designed a continual learning approach for lane detection, reaching high accuracy using just **41%** of parameters.
 - Created test-time architecture for few-shot semantic segmentation of lanes using COCO/PASCAL-pretrained models, achieving a **+433.82%** improvement over baseline.
 - Leveraged large-scale vision foundation models to detect **statistically significant covariate shifts** in road environments across the CULane, CurveLanes, AssistTaxi, and Waymo Open Dataset.
- Microsoft-Volunteer Research** Spring 2025
Guide: Yash Jain, Vibhav Vineet, Siddhartha Bhattacharyya **Microsoft Research, USA**
 - Topics: Multi-modal Learning (Text and Vision), Vision Foundation Models, Prompt Optimization
 - Developed a closed-loop test-time prompt refinement framework using multimodal LLMs, improving text-to-image generation by **+20.34%** on DALL-E 3, **+22.75%** on Flux, and upto **+39.36%** on Stable Diffusion models.
- Defense Advanced Research Projects Agency (DARPA)-Funded Research** Summer 2025
Guide: Junaid Babar, Isaac Amundson, Siddhartha Bhattacharyya **Collins Aerospace, USA**
 - Topics: Formal Verification, Reinforcement Learning, Model Checking, Language Translation
 - Built an automated Soar-to-PRISM translator in ANTLR/Java to enable formal verification of cognitive models.
- AHA-Funded Research** Summer 2024
Guide: Venkat Keshav Chivukula **UTHealth Houston, USA**
 - Topics: Self-Supervised Learning, Representation Learning, 3D/4D Reconstruction, Computational Fluid Dynamics
 - Designed LVADNet3D, an autoencoder for intraventricular velocity reconstruction across x, y, and z directions, outperforming UNet3D with **34.13%** lower error and **10.55%** higher fidelity.
 - Generated synthetic CFD hemodynamic datasets of **250 LVAD patients** for training 3D deep learning models.
- Software Engineer Intern at Sentient Labs** Summer 2023
Guide: Anshul Singhal, Nilesch Goel **Sentient Labs, UAE**

- Built fleet management and health monitoring solutions for **50+** commercial drones and leveraged AWS IoT Services.
- Applied causal inference to assess how usage patterns affected drone uptime and cleaning efficiency.

AWARDS

- Awarded for **Outstanding Academic Achievement** at Florida Tech 2025
- Outstanding Undergraduate Thesis Award** for excellent undergraduate thesis at BITS Pilani Dubai 2023
- Winners in IEEE Robotics Competition**, for project **3D Object Detection and Tracking for Road Safety** 2023

SKILLS

Programming Languages: Python, C/C++, Java, R, SQL

ML Frameworks & Libraries: PyTorch, TensorFlow, JAX, sklearn, rLLm, VeRL, Diffusers, MMDetection, OpenCV

Cloud Platforms & Services: Azure (ML Studio, DevOps), AWS (S3, SageMaker), GCP (Google Maps, Cloud Storage),

PUBLICATIONS

* equal contribution

Adapt, But Don't Forget: Fine-Tuning and Contrastive Routing for Lane Detection under Distribution Shift [\[PDF\]](#)

[M.A. Hafeez Khan](#), Parth Ganeriwala, Sarah M. Lehman, Siddhartha Bhattacharyya, Amy Alvarez, Natasha Neogi
Workshop on Out Of Label Hazard Detection in Autonomous Driving at **ICCV**, 2025

Test-time Prompt Refinement for Text-to-Image Models [\[PDF\]](#)

[M.A. Hafeez Khan](#)*, Yash Jain*, Siddhartha Bhattacharyya, Vibhav Vineet

[\[Invited Talk\]](#) Workshop on Multimodal Reasoning and Slow Thinking in Large Model Era at **ICCV**, 2025

Modular Test-time Input-Space Refinement for Few-Shot Segmentation

[M.A. Hafeez Khan](#), Parth Ganeriwala, Amy Alvarez, Siddhartha Bhattacharyya

Under submission at NeurIPS conference 2025

Few-Shot Classification and Anatomical Localization of Tissues in SPECT Imaging [\[PDF\]](#)

[M.A. Hafeez Khan](#), Samuel M. Boddepalli, Siddhartha Bhattacharyya, Debasis Mitra

[\[Oral Presentation\]](#) Medical Imaging Conference (**MIC**), 2025

LVADNet3D: A Deep Autoencoder for Reconstructing 3D Intraventricular Flow from Sparse Data

[M.A. Hafeez Khan](#), Marcello Mattei, Ben Diaz, Ruth White, Siddhartha Bhattacharyya, Venkat Keshav Chivukula

[\[Oral Presentation\]](#) International Conference on Machine Learning and Applications (**ICMLA**), 2025

NORA: A Nephrology-Oriented Representation Learning Approach Towards CKD Classification

[M.A. Hafeez Khan](#), T. Bhattacharyya, O. Khan, Noorah Khan, Alina Khan, M.Q. Khan, Sujoy Hajra

[\[Spotlight Paper\]](#) International Conference on Machine Learning and Applications (**ICMLA**), 2025

ALINA: Advanced Line Identification and Notation Algorithm [\[PDF\]](#)

[M.A. Hafeez Khan](#), Parth Ganeriwala, Siddhartha Bhattacharyya, Natasha Neogi, Raja Muthalagu

IEEE / CVF Computer Vision and Pattern Recognition Conference **CVPR**, 2024

A Hybrid BiLSTM-CNN Approach for Intrusion Detection for IoT Applications [\[PDF\]](#)

[M.A. Hafeez Khan](#), Sapna Sadhwani, Raja Muthalagu, Pranav Pawar, K. Suresh

Scientific Reports, Springer, 2024

AssistTaxi: A Comprehensive Dataset for Taxiway Analysis and Autonomous Operations [\[PDF\]](#)

Parth Ganeriwala, Siddhartha Bhattacharyya, S. Gunther, Brian Kish, [M.A. Hafeez Khan](#), Aknur Dhadoti, Natasha Neogi

International Conference on Machine Learning and Applications (**ICMLA**), 2023

Classification of Microstructure Images of Metals Using Transfer Learning [\[PDF\]](#)

[M.A. Hafeez Khan](#), Hrishikesh Sabnis, J. Angel Arul Jothi, J. Kanishkha, A.D. Prasad

International Conference on Modelling and Development of Intelligent Systems (**MDIS**), 2022

Detection of Cavities from Oral Images using Convolutional Neural Networks [\[PDF\]](#)

[M.A. Hafeez Khan](#), Giri Prasad S., J. Angel Arul Jothi

[\[Best Paper Award\]](#) IEEE International Conference on Electrical, Computer and Energy Technologies (**ICECET**), 2022

Detection of Bicep Form Using Myoware and Machine Learning [\[PDF\]](#)

[M.A. Hafeez Khan](#), Rohan V. Rudraraju, R. Swarnalatha

International Conference on Advances in Data-driven Computing and Intelligent Systems (**ADCIS**), 2022

TEACHING EXPERIENCE

Graduate Teaching Assistant

Florida Tech

Primary Instructor: Prof. Philip Chan (FLTech)

Aug '23 – May '24

- Course: CSE2010 Algorithms & Data Structures; My Rating: 4.9/5.